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Department of Information Science and Engineering

"IoT Based Cake Freshness and Adulteration Detection Device"

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Agenda

- Introduction to the project
- Problem Statement
- Scope of the Project
- Existing System
- Proposed System
- Conclusion

Introduction

- Food safety important for human health
- Cakes spoil due to microbial contamination.
- Stored cakes sensitive to environmental conditions.
- Expiry date not always reliable indicator.
- Early cake spoilage invisible to consumers
- Sensors help monitor cake quality automatically

Problem Statement

Cake spoilage and adulteration are major concerns in food safety. Cakes can spoil due to microbial growth, improper storage conditions, or the presence of harmful additives.

In many cases, early spoilage in cakes cannot be easily detected by consumers through visual inspection or smell. Traditional methods for detecting food quality mainly involve laboratory testing, which requires specialized equipment, trained personnel, and significant time. These methods are not practical for everyday use in cake shops or households. Therefore, there is a need for a simple, portable, and cost-effective system that can detect cake freshness and possible adulteration quickly using sensor-based technology.

Scope of the Project

- Focuses on unpackaged cake quality monitoring
- Helps identify unsafe cake conditions early
- Reduces dependency on laboratory testing
- Supports low-cost food safety monitoring
- Designed for simple daily cake inspection
- Suitable for bakery and shop environment

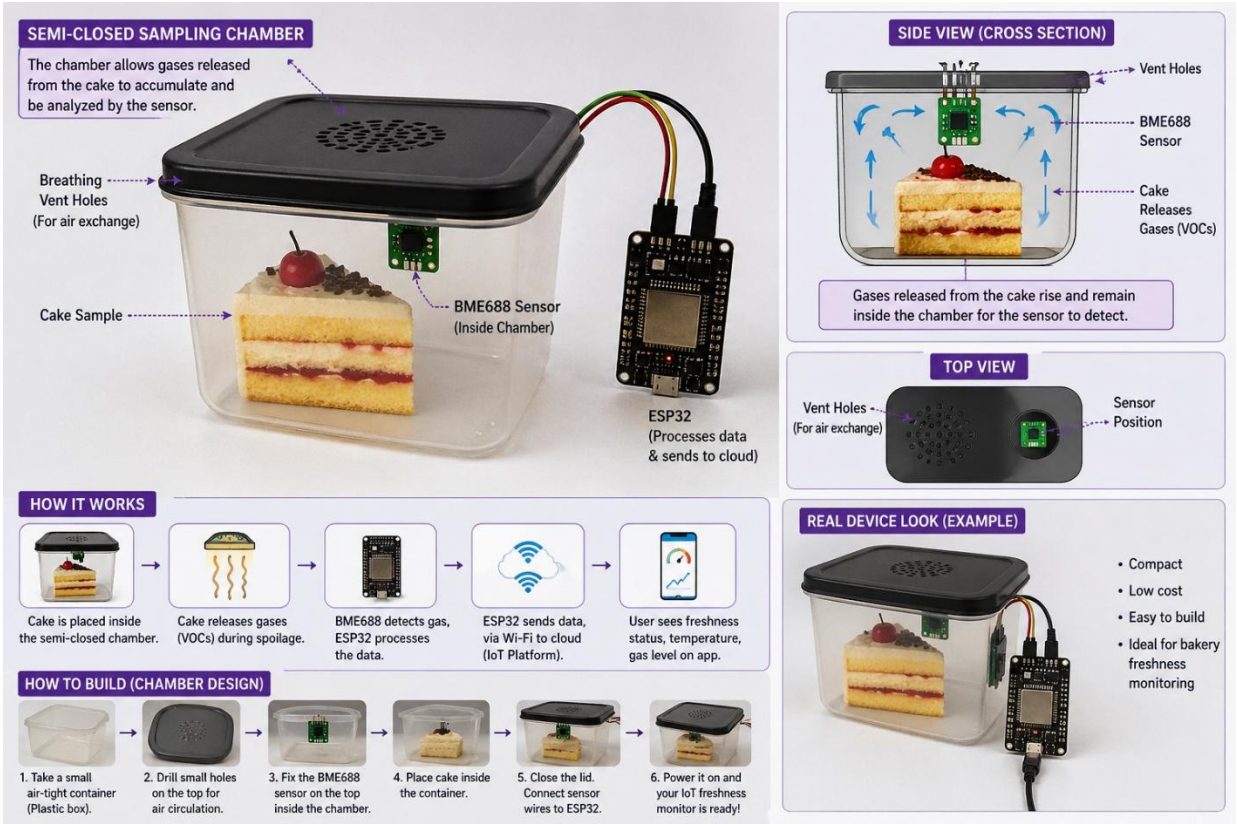
Existing System

- Cake quality testing done laboratory methods
- Methods include chemical microbial analysis techniques.
- Monitor environmental temperature humidity conditions
- Requires expensive equipment trained professionals.
- Not suitable for daily cake inspection
- Difficult for small cake businesses



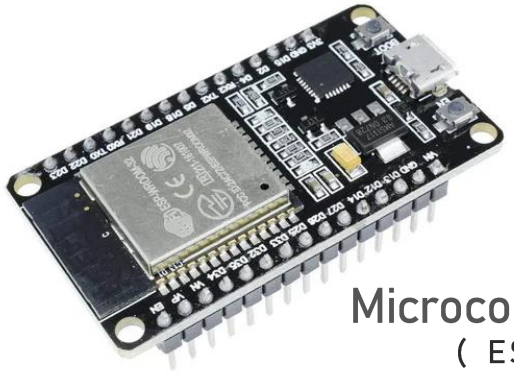
Proposed System

- Smart device detects cake freshness automatically.
- Sensors analyze different cake quality parameters.
- Gas sensor detects cake spoilage gases.
- Temperature humidity sensor monitors environment conditions.
- Color sensor identifies abnormal cake color.
- Microcontroller processes data shows freshness result.





Gas Sensor
(BME688)



Microcontroller
(ESP32)

Conclusion

The proposed IoT-based bakery freshness monitoring system provides a smart and low-cost solution for detecting spoilage in cakes and bakery products using the ESP32 WROOM NodeMCU Module with 30 Pins and Bosch BME688 AI Environmental Sensor. The system monitors spoilage-related gases and displays freshness status through mobile connectivity using Wi-Fi and Bluetooth. This helps improve food safety and reduce bakery product wastage.

Future Enhancements

In future, the system can be enhanced by integrating a rechargeable battery module to eliminate the need for an external power bank and make the device fully portable. Additional improvements may include AI-based spoilage prediction, cloud analytics, smart mobile notifications, and support for multiple bakery product classifications. The device can also be upgraded into a compact commercial IoT product for real-time smart bakery monitoring applications.

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Thank You